

Title: Paradigm uniformity is probabilistic: Evidence from velar nasalization in Japanese
[NOTE: talk preferred, but able to present both as a poster or as a talk]

Synopsis: Recent work by Steriade & Stanton (2020) and Breiss (2020, 2021) suggests that Lexical Conservatism, a type of “base” effect where the phonological grammar references the contents of the lexicon, can be probabilistic. This paper offers further evidence for the probabilistic nature of base effects with evidence from another type of base effect, paradigm uniformity, based on examination of voiced velar nasalization in the Yamanote dialect of Japanese. We also demonstrate that the probabilistic paradigm uniformity is modulated by the frequency of the non-local output form (the remote base). We propose a formal model which integrates the frequency effect using a scale of faithfulness constraints as proposed in Coetzee & Kawahara (2013) - henceforth C&K - indexed to the output-output faithfulness constraint.

Voiced Velar Nasalization: In various dialects of Japanese, [g] and [ŋ] stand in an allophonic relationship, with [g] occurring in prosodic-word-initial position, and [ŋ] occurring elsewhere (e.g. [gama] “toad” vs. [kaŋami] “mirror”). The relevance of this pattern to theoretical phonology was made clear by Ito & Mester (1997), which the current study substantially builds upon. The pattern of velar nasalization is sensitive to a range of morphophonological factors (Breiss et al. 2021), but for this talk the relevant generalizations are the following:

- (1) In compounds, when the second member (=N2) begins with [g] and is a free morpheme, voiced velar nasalization targeting that [g] is optional.
(a) E.g. N2=/ga/ “moth”, [doku-**ga**]~[doku-**ŋa**] “poison moth”
- (2) When N2 is a bound morpheme, voiced velar nasalization is obligatory.
(b) E.g. N2=/ga/ “fang”, [doku-**ŋa**], *[doku-**ga**] “poison fang”
- (3) When N2 begins with [k], which is rendered [g] due to an independent voicing process (Rendaku), voiced velar nasalization is obligatory.
(c) E.g. N2=/kaki/ “writing”, [joko-**ŋaki**], *[joko-**gaki**] “horizontal writing”

(1) shows that voiced velar nasalization is not merely a static generalization about surface allophonic patterns, but it in addition manifests itself as an active (morpho)phonological alternation. Ito & Mester (1997) proposes that the optionality in (1) arises from the conflict between a force requiring VVN and output-output correspondence between the free [g]-initial form of N2 --- the remote base --- and its status in the compound. A consequence of this argument is that when N2 is bound, as in (2), output-output correspondence is irrelevant because there is no free [g]-initial form of N2, and thus VVN is obligatory. (3) implies a feeding process between Rendaku and VVN, and results in a saltatory phonological alternation (Hayes & White 2015): underlying /k/ is realized as [ŋ], skipping the phonetically intermediate [g], whereas underlying /g/ at least sometimes remains [g] by output-output correspondence to a free N2. This saltatory pattern makes sense in terms of Ito & Mester’s (1997) proposal: since the free base of N2 begins with [k], not [g], output-output correspondence could not require that the bound form of N2 in compounds begin with [g].

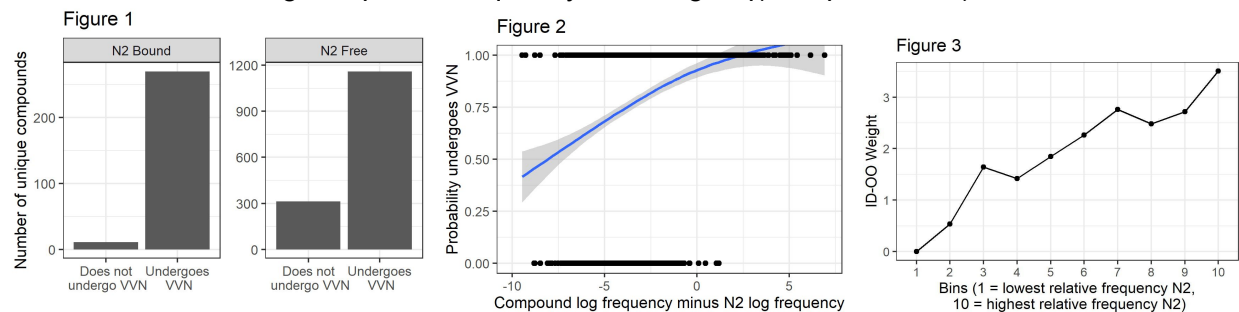
Research aims: Our paper explores these generalizations quantitatively, and proposes a model using a MaxEnt grammar which integrates the role of frequency of the N2’s free form in conditioning the variation in VVN.

Methods: We extracted all compound words that had /g/-initial N2s from the *NHK Pronunciation and Accent Dictionary*, and recorded their listed pronunciations (having undergone VVN, as in [hai-**ŋan**] “lung cancer”, or not, as in [noo-**geka**] “brain surgery”). In cases where two pronunciations were listed, we recorded the pronunciation listed as preferred or predominant. We then used the Balanced Corpus of Contemporary Written Japanese (BCCWJ) to find

part-of-speech tags for N1 and N2 in each compound. We then excluded all compounds which contained either N1 or N2 with a non-noun tag, leaving 1,329 compounds with a /g/-initial N2. Finally, we used BCCWJ to annotate each compound with the log-frequency of the compound and N2. In cases where a form was absent from BCCWJ, we gave a log-frequency of 0. We also extracted all compounds with /k/-initial N2s not containing a voiced obstruent, but did not annotate them further because they categorically undergo VVN.

Results: We first assessed the evidence for the observations proposed by Ito & Mester, (1)-(3) above. Figure 1 depicts the proportion of compounds undergoing VVN, depending on the status of the N2. The vast majority of compounds with bound N2s undergo VVN (left panel), supporting (2). Those with free N2s (right panel) exhibit substantially more variation, supporting (1). We also found that compounds with /k/-initial N2s which undergo Rendaku categorically undergo VVN as well, supporting point (3) (not pictured here for sake of space).

The fact that there is a substantial amount of variation in VVN among compounds with free N2s (right panel of Figure 1) suggests that paradigm uniformity is probabilistic. Moreover, we find that this variation is conditioned by the relative frequencies of the compound and N2 (Figure 2). This finding is verified using a Bayesian logistic regression model fit using *brms* (Bürkner, 2017), with fixed effects of log compound frequency and $\Delta \log \text{freq}(\text{Compound}, \text{N2})$



Theoretical proposal: We adapt C&K's proposal that the weights of faithfulness constraints can be scaled according to lexical frequency. We also follow C&K's method in modelling the frequency effect in compounds with free N2s by "binning" compounds by the relative frequency of the compound and N2 into deciles, and taking the average probability of undergoing VVN per bin as an input, alongside the compounds with bound N2, and those with /k/-initial N2 inputs to be modeled. We then fit a MaxEnt harmonic grammar (Smolensky 1986, Goldwater & Johnson 2003) with the following constraints, cleaving closely to Ito & Mester's analysis: *VgV, * η , ID-IO, RENDAKU, and one ID-OO constraint for each frequency bin. Figure 3 plots the weight of ID-OO (vertical axis) as a function of its bin (horizontal axis). We find that the weight of ID-OO decreases with the bin denoting relative frequency of N2. This makes intuitive sense: the more salient the free N2 is in the mind of the speaker, the more likely faithfulness to that form will influence compound realization.

Discussion: We provide an analysis of demonstrated probabilistic paradigm uniformity in VVN, incorporating both standard phonological markedness-avoidance effects and frequency-regulated faithfulness which resists the markedness-driven alternation. It is important to note that the frequency effect of free /g/-initial N2s cannot simply be attributed to an extra-grammatical, exemplar-like tendency for a morpheme to resemble its free form in proportion form to the strength of encoding of its free form in the lexicon: critically, Rendaku-eligible /k/-initial free N2s categorically undergo VVN, regardless of their relative frequency. This demonstrates that the probabilistic paradigm uniformity, and specifically its frequency-conditioning, must be modeled as part of the phonological grammar.